

AVIKAL PANDEY

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OVERVIEW

B.Tech undergraduate specializing in **backend development** and scalable system design. Experienced in building applications using **Java, REST APIs, and distributed systems**, with strong foundations in **Data Structures and Algorithms**. Focused on designing efficient, production-ready systems with clean architecture.

EDUCATION

JSS Academy of Technical Education, Noida	India
B.Tech in Computer Science	2023 – 2027

Dr. Virendra Swarup Education Center, Kanpur	2021 – 2022
Class XII – 96%	

TECHNICAL SKILLS

- **Languages:** Java, C++, JavaScript, Python
- **Backend:** Spring Boot, FastAPI, RESTful APIs
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Frontend:** HTML, CSS, JavaScript, Tailwind CSS
- **Tools:** Git, GitHub, Postman, Docker, Vercel, Render
- **Core:** Data Structures & Algorithms, OOP, Computer Networks, System Design

PROJECTS

- **ULTRON** *AI Assistant System*
 - Developed an AI-powered assistant in Python featuring hotword detection, speech/text I/O, and Open-Router API integration
 - Attained sub-2s response times by streamlining API interactions and constructing lightweight processing pipelines
 - Devised an offline-capable GUI to guarantee core functionalities independent of network connectivity
 - Designed a modular, plugin-based framework to support extensible automation workflows
- **BARTR** *Product & Skill Exchange Platform*
 - Orchestrated a full-stack exchange platform facilitating seamless user listings, browsing, and peer-to-peer skill trades
 - Structured scalable data models and intuitive interaction flows for organized listing management
 - Slashed data fetch latency by ~30% through fine-tuned database queries and efficient state-handling mechanisms
 - Directed end-to-end application deployment while managing the complete product lifecycle
- **TerraNest (In Progress)** *Property Livability Predictor*
 - Model scalable backend architectures tailored for high-volume property listings and transactional workflows
 - Formulate robust RESTful APIs and normalized database schemas optimized for multi-user operations
 - Deploy advanced filtering, pagination, and query-optimization strategies to minimize database overhead
 - Champion clean, modular design principles to guarantee long-term system scalability and performance

EXPERIENCE

- **Software Engineering Job Simulation** *Telstra*
 - Implemented 5+ RESTful API endpoints in Java to streamline CRUD operations across relational databases
 - Refactored application architecture using clean design principles, reducing code redundancy by 20%
- **Software Engineering Job Simulation** *JP Morgan Chase & Co.*
 - Architected 4 Spring Boot REST controllers to efficiently process mock transactional data streams
 - Integrated an in-memory H2 database with optimized queries, achieving sub-10ms data retrieval latencies